

CSS — Lecture 09

Coding and Web Development | Summary Notes

1. What is CSS?

CSS stands for Cascading Style Sheet. It is used for formatting and styling the appearance of a webpage.

HTML vs CSS — a simple analogy:

- HTML is like the structure (skeleton) of a house — it defines the basic parts and arrangement (walls, doors, windows, roof).
- CSS is like the style/design of the house — it adds paint, furniture, lighting, and decoration to make it beautiful.

In short: HTML builds the structure, and CSS styles and beautifies it. Together, they build a complete webpage.

2. Types of CSS

There are 3 types of CSS: Inline CSS, Internal CSS, and External CSS.

2.1 Inline CSS

Inline CSS is used to apply a unique style to a single HTML element by using the style attribute directly within the HTML tag.

```
<p style="color: red;">Hello</p>
```

2.2 Internal CSS

Internal CSS (also called embedded CSS) applies styles to a single HTML page by placing all the CSS code inside a <style> tag within the <head> section of the document.

```
<head>
  <style>
    p { color: red; }
  </style>
</head>
```

2.3 External CSS

External CSS stores style rules in a completely separate file with a .css extension, and links it to the HTML document using the <link> tag.

```
<link rel="stylesheet" href="index.css">
```

Here, index.css is the external stylesheet file, and the <link> tag inside the HTML file's <head> connects it to the HTML page.

3. CSS Selectors

CSS selectors are patterns used to target and select specific HTML elements to style. The basic syntax is:

```
selector {
  property: value;
}
```

3.1 Element (Type) Selector

Selects HTML elements directly by their tag name.

```
p {  
  color: red;  
}
```

3.2 Class Selector

Targets HTML elements that have a specific class attribute, letting you apply identical styling to multiple elements. Written with a period (.) followed by the class name.

```
<p class="hello">Hello</p>  
  
.hello {  
  color: blue;  
}
```

3.3 ID Selector

Matches and styles a single, unique HTML element based on its id attribute (id should be unique on the page). Written with a hash (#) followed by the id name.

```
<p id="sagar">Hello</p>  
  
#sagar {  
  color: green;  
}
```

3.4 Universal Selector

The universal selector (*) matches every single HTML element on the page — it acts as a wildcard, applying styles globally to all tags.

```
* {  
  margin: 0;  
  padding: 0;  
}
```

4. CSS Specificity

Specificity is the algorithm browsers use to decide which style rule wins when multiple rules target the same element. Higher specificity overrides lower specificity.

Order of specificity (highest to lowest):

Inline CSS > ID Selector > Class Selector > Element Selector

5. CSS Combinators

Combinators define the relationship between two selectors — describing how elements relate to each other in the HTML structure (e.g., parent, child, siblings).

Combinator	Symbol	Meaning
Descendant	A B	B is nested inside A, at any depth
Child	A > B	B is a direct (immediate) child of A
Adjacent Sibling	A + B	B immediately follows A, same parent
General Sibling	A ~ B	B is any sibling appearing after A, same parent

Descendant Combinator (space)

Selects all elements nested inside a specified ancestor element, no matter how deep.

```
div p {  
  color: red;  
}
```

Child Combinator (>)

Targets only the direct/immediate children of a specified parent element.

```
div > p {  
  color: red;  
}
```

Adjacent Sibling Combinator (+)

Matches the second element only if it comes immediately after the first, sharing the same parent.

```
h1 + p {  
  color: red;  
}
```

General Sibling Combinator (~)

Matches all instances of the second element that come after the first element, sharing the same parent.

```
h1 ~ p {  
  color: red;  
}
```

6. Pseudo-classes & Pseudo-elements

6.1 Pseudo-class

A pseudo-class is a keyword added to a selector that defines a special state of the selected element (e.g., when hovered, or its position among siblings).

```
selector:pseudo-class {  
  property: value;  
}
```

Example — nth-child formulas: $2n$ (even elements), $2n+1$ (odd elements).

6.2 Pseudo-element

A pseudo-element is a keyword added to a selector that lets you style a specific part/content of an HTML element.

```
selector::pseudo-element {  
  property: value;  
}
```

6.3 Pseudo-class vs Pseudo-element

	Pseudo-class	Pseudo-element
Syntax	Single colon :	Double colon ::
Targets	Element's state	Element's part / content
Example	:hover, :nth-child	::before, ::after